Accenture — Aptitude Test 4

Questions 76–100 of 100 · 25 questions

Question 76. Three runners A, B and C run a race, with runner A finishing 12 meters ahead of runner B and 18 meters ahead of runner C, while runner B finishes 8 meters ahead of runner C. Each runner travels the entire distance at a constant speed. What was the length of the race?

Question 77. In a stream running at 2kmph, a motor boat goes 6km upstream and back again to the starting point in 33 minutes. Find the speed of the motor boat in still water.

Solution:
Time taken in terms oh hours = 33/60
Total time
= upstream time + downstream time
33/60 = [6/(Sb – Sr)] + [6/(Sb – Sr)]
33/360 = [1/(Sb – 2)] + [1/(Sb + 2)]
11/120 = [(Sb + 2 + Sb – 2)/(Sb

Question 78. Riya and Priya set on a journey from same point. Riya moves eastward at a speed of 20kmph and Priya moves westward at a speed of 30 kmph. How far will be Priya from Riya after 30 minutes?

Question 79. There are three runners Tom, Dick and Harry with their respective speeds of 10kmph, 20kmph and 30kmph they are initially at P and they have to run between the two points P and Q which are 10 km apart from each other. They start their race at 6 am and end at 6 pm on the same day. If they run between P and Q without any break, then how many times they will be together either at P or Q during the given time period?

Solution:
Time taken for Tom to travel 20km
= 20/10 = 2 hours = 120 minutes
Time taken for Dick to travel 20 km
= 20/20 = 1hour = 60 minutes
Time taken for Harry to travel 20 km
= 20/30 = 0.66 hours = 40 minutes.

LCM of (120, 60, 40) = 120
In every 2 hours the three persons will meet.
In 12 hours they will meet for 12/2 = 6 times.

Question 80. Robert is travelling on his cycle and has calculated to reach point A at 2 P.M. if he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed must he travel to reach A at 1 P.M.?

Solution:
If a person increase the speed by ½ then the time taken will reduce by 1/3rd .
Here Robert increases the speed by ½ and the time taken reduced by 2 hours.
2 hours = (1/3) x Time taken at 10kmph
Time taken = 6 hours.
Distance = 6 x 10 = 60 kmph.
He started his journey 6 hours before 2pm i.e. at 8am.
Time taken from 8am to 1pm = 5 hours
Speed required to cover 60km in 5 hours = 60/5 = 12 kmph.

Question 81. A man rows to a place 48 km distant and come back in 14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream.

What is the rate of the stream?

Solution:
Speed is directly proportion to distance
Speed ratio = 3 : 4
Time is inversely proportional to speed.
Time ratio = 4 : 3
Assume that the actual times = 4x and 3x
4x + 3x = 7x = 14 hours ■x = 2
Actual time taken upstream and downstream = 8 hours , 6 hours
SUS, Upstream speed = 48/8 = 6kmph
SDS, Downstream speed=48/6 = 8 kmph
Speed of current = (SDS – SUS)/2 = 1 kmph

Question 82. A supportive young hare and tortoise raced in opposite directions around a

circular track that was 100 km in diameter. They started at the same spot, but the hare did not move until the tortoise had a start of one eighth of the distance (that is, the circumference of the circle). The hare held such a poor opinion of the other's racing ability that he sauntered along, nibbling the grass until he met the tortoise. At this point the hare had gone one sixth of the distance. How many times faster than he went before must the hare now run in order to win the race?

Solution:
The circumference value can be anything for this question. So, let us take a number which is a multiple of both 6 and 8 for easy calculation.
Assume that the race track is 48 km long.
Hare gives 6 km start to tortoise.
Assume that the hare and tortoise meet at 1/6th the distance in 1 hour.
Distance travelled by hare = 8 km and distance travelled by tortoise = 34 yards
Initial speed of hare = 8 kmph
Speed of tortoise = 34kmph.
Time taken for tortoise to cover the 8 km distance = 34/8 = 4.25 hours.
To win the race the hare must reach the

A c c e n t u r e

Solution:
Total distance to be travelled to cross each other = 2km + 200m + 300m
= 2500m
Time taken = 2500/(20 + 30) = 50 sec.

end point within 4.25 hours.
Distance to be travelled by hare within 4.25 hours = 40 km.
Speed of hare must be greater than
= 40/4.25 = 9.41 kmph.
Fractional increase from the initial speed of the hare to required speed
= (9.41 – 8)/8 = 0.176 or 17.6 %.

Question 83. A train 120 meter long passes an electric pole in 12 seconds and another train of

**same
length
traveling
in
opposite direction in 8 seconds. The
speed of the second train is?**

Solution:
Total time = 4.5 hours
Upstream time+Downstream time = 4.5
4.5 = [30/(15 + Sr)] + [30/(15 – Sr)]
3/20 = [(15 – Sr) + (15+Sr)]/(225–Sr

2)
3/20 = 30/(225 – Sr

Solution:
Let the speed of the trains be Sa and Sb
Time taken to cross a pole
= 120/Sa = 12
Sa = 10 mps.
Speed of the first train = 10mps.
Time taken to cross another train
= [(120+120)/(Sa + Sb)] = 8
240 = 8 x 10 + 8Sb
8Sb = 160
Speed of second train = 20 mps.

2)
675 – 3Sr

2 = 600
3Sr

Question 84. Two trains, 200 and 160 meters long take a minute to cross each other while traveling in the same direction and take only 10 seconds when they cross in opposite directions. What are the speeds at which the trains are traveling?

Question 85. Two trains are 2km apart. Speed of one train is 20m/s and the other train is running at 30 m/s. Lengths of the trains are 200 and 300m. In how much time do the trains cross each other?

2 = 75 ■Sr = 5 kmph

Question 86. A boat whose speed is 15 kmph in still water goes 30 kmph downwards and come back in total 4 hours 30 min. What is the speed of stream in kmph?

Question 87. How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

Solution:
The last digit must be 5.
The remaining 2 digits for the three digit number can be arranged from the remaining 5 digits.
Number of ways = 5P2 = 20

Question 88. How many ways to arrange a word ORANGE in which vowels are not together?

Solution:
In same direction:
60 = [(200 + 160)/(Sa - Sb)]
Sa - Sb = 6
---(1)
In opposite direction:
10 = [(200 + 160)/(Sa + Sb)]
Sa + Sb = 36
---(2)
By solving (1) and (2), we get
Sa = 21 and Sb = 15

Solution:
Consider the three vowels in the word orange as a single element.

The number of elements now is 4 which can be arranged in 4! ways.
The three vowels which are together can be arranged in 3! ways.
Number of arrangements in which vowels are together = 3! X 4!
= 144.
Total arrangements
= 6! = 720.
Vowels not together
= 720 – 144 = 576

Question 89. In how many ways can a lock be opened if that lock has three digit number lock if sum of the first two digits is less than or equal to the last digit? Numbers are from 0-9.

Solution:
The 3rd digit = 9
If the sum of 1st and 2nd digit = 9, the results are (0,9), (9,0), (1,8), (8,1), (2,7), (7,2), (3,6), (6,3), (4,5), (5,4) = 10 results.
If the sum = 8, the results are (0,8), (8,0), (1,7), (7,1), (2,6), (6,2), (3,5), (5,3), (4,4) = 9 results.
The number of results will decrease by 1 for the successive sums.
Total results = 10+9+8+…+2+1 = 55

Question 90. A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw?

Solution:
Total ways of selecting 3 balls – ways of selecting 3 balls from white and red balls = ways of selecting 3 balls with at least 1 black ball.
Number of selections = 9C3 – 6C3 = 104.

Question 91. The sides AB, BC and CA of a triangle ABC having 3, 4 and 5 interior points receptively on them. The total number of triangles that can be constructed by using these points as vertices is?

Solution:

Selecting any three points – selecting three points from the same line = Number of triangles.
Total number of points = 3+4+5 = 12
12C3 – 3C3 – 4C3 – 5C3 = 220 – 15 = 205 triangles can be formed.

Question 92. After a get-together every person present shakes the hand of every other person. If there were 105 hands-shakes in all, how many persons were present in the party?

Solutions:
Number of handshakes = nC2, because there are two persons involved in a handshake and 1st man shaking hand with 2nd man is not different from 2nd

man with the 1st man.
nC2 = n(n – 1)/(2 x 1)= 105
n(n – 1) = 210
Product of two successive numbers=210
By solving we get, n = 15
15 persons attended the party.

Question 93. In how many ways a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?

Solution:
8C3 x 10C6 = 8C3 x 10C4 = 56x210= 11760

Question 94. A class consists of 100 students, 25 of them are girls and 75 boys; 20 of them are rich and remaining poor; 40 of them are fair complexioned. The probability of selecting a fair complexioned rich girl is?

Solution:
Probability of selecting a girl = ¼
Probability of selecting a rich person
= 1/5
Probability of selecting fair complexion
= 2/5
Overall probability = (1/4)(1/5)(2/5)
= 2/100 = 1/50

Question 95. A box contains 5 brown and 4 white socks. A man takes out two socks. The probability that they are of the same color is?

A c c e n t u r e

Solution:
The 2 socks may be either brown or white.
Selections = 5C2 + 4C2 = 16
Probability = 16/[9C2] = 16/36 = 4/9

Question 96. A coin is tossed and simultaneously a dice is rolled. What is the probability that the coin will show heads an the dice will have a composite number?

There must be a carry over for M.
O + C = E, that means 0 + C = E.
So, there must be a carry over for W + H

Solution:
Probability for head = 1/2
Probability for composite number = 3/6
Overall probability
= (1/2) x (1/2) = 1/4

Question 97. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red are?

Solution:
Ways of selecting 3 red balls = 5C3 = 10
Ways of selecting any 3 balls=15C3 = 455
Probability = 10/455

Question 98. Find the probability that a leap year contains 53 Sundays.

Solution:
Reminder of 366/7 = 2
A leap year has two odd days.
If the leap year starts with Sunday, it will end in Monday.
If it starts with Saturday, it will end in Sunday.
Out of 7 days in a week, if the leap year starts on these two days, we will have 53 Sundays.
Probability = 2/7

P = 1; O = 0; W = 5; E = 4; R = 2
P+O+W+E+R = 1+0+5+4+2 = 12

Question 99. If f(x) = ■(x² – 50)■, what is the value of f(-5)?

Solution:
F(-5) = ■(-52 – 50) ■= 25.

Question 100. HOW+MUCH=POWER. Find the value of P+O+W+E+R

A c c e n t u r e

Solution:
The question is from Crypt Arithmetic.
A three digit number is added to a four digit number to get a five digit number.
So,
P = 1; O = 0 and M = 9

1
1
H
O
W
+ M U

C
H
P O W E
R

P O M W H R U C E

1 0
9
2
9
X

1 0
9
2
8
0
X

1 0
9
3
9
X

1 0
9
3
8
1
X

1 0
9
3
7
0
X

1 0
9
4
9
X

1 0
9
4
8
2 4
X

1 0
9
4
7
1
X

1 0
9

5
9
X

1 0
9
5
8
3 5
X

1 0
9
5
7
2 8 3 4 √

The name Accenture

is an acronym for

“Accent on the

Future”

Accent (v) - Emphasize